An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

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1. Executive summary

Floods displace millions of people in India every year. The slow part of a rescue is not reaching survivors — it is *finding* them.

We are building three one-metre, 6.4 kg multirotor aircraft that search flooded ground as a coordinated group under **one operator**, recognise people from the air, report their position to about a metre, and drop a relief kit to each. It needs **no mobile network and no internet** — among the first things a flood removes.

The sizing, error budgets and flight software are written down and reproducible, and coordination has been demonstrated across three simulated autopilots. **No aircraft has flown yet.** While checking our own arithmetic we also found a real flaw: the camera sees a survivor clearly and our software discards that detail before the detector sees it. We have a fix and can test it on free public data first.

We are asking for **INR 7,34,379 across 32 steps**, none above INR 30,000, each released only on the result of the one before. The first is INR 26,196.

2. The problem

After a flood, the binding constraint is time-to-locate. Boat teams search slowly and at high risk and cannot see onto rooftops. Helicopters are scarce, costly per hour, and committed to evacuation. Nothing fills the gap between them: covering a large area quickly and reporting exactly where people are.

Small drones look like the answer and mostly are not: they need one pilot each, depend on a network that the disaster has removed, find people without helping them — and seeing a person from the air is genuinely hard.

That last part is where we found our own mistake. A person in floodwater shows head and shoulders only, about 40 cm across. Our camera at 60 m resolves that across 26 pixels; but detectors take a fixed small input, so the image is shrunk and the person becomes **13 pixels**. Published results at that size report 29–61 % accuracy. Choosing a smaller, faster model makes it *worse*, because smaller models take smaller inputs.

Three things make floods harder still: no flood dataset labels *people*; thermal fails in water, where skin cools to water temperature; and we had been sizing for a person lying flat (1.7 m) rather than one in water (0.4 m), which made the problem look four times easier than it is. We found that ourselves, and it is why we now want to measure rather than assume.

3. Our solution

The system. The operator draws a search area of any shape; the software divides it into three balanced pieces and plans a path over each — verified on thirteen awkward shapes with no path leaving the region and under 0.02 % imbalance. The aircraft then fly themselves: our harness proves structurally that no command is sent after the mission starts. Detection and geolocation run *on board*, so nothing depends on a radio link. Each aircraft carries four kits and can descend, stop over a survivor and release one. Link loss, low battery and operator abort all converge on the same return-and-land behaviour.

The fix for the vision flaw, in order of effect:

1. **Fly lower.** 60 m to 30 m more than triples the target's size, and the mission takes under eight minutes so we have time to spare.

2. **Stop shrinking the picture.** Cut the full-resolution image into overlapping tiles instead. Standard practice; costs four times the computing.
3. **Afford that with a cheap first look.** Most of a flood is empty water. A small fast model asks only *"could anything be here?"* and the expensive detector runs where it says yes. About 87 % of tiles are empty and we need to skip 80 % to break even, so the arithmetic closes.

Steps 1 and 2 take a person from **13 pixels to 52** — sixteen times the area — for the same computing cost.

The honest risk: the cheap first look could discard a tile containing a person. That failure is silent from the air and cannot be undone, which is why we test before adopting, and why our pass mark assumes such mistakes repeat across frames rather than cancelling out.

4. Deliverables

1. Three aircraft flying as a coordinated group under one operator.
2. A ground station that shows everything and is deliberately *unable* to fly the aircraft, so the autonomy cannot be quietly undermined.
3. Autonomy software — area division, path planning, allocation, failsafes — with an automated test suite.
4. A *measured* detection figure, replacing the calculated one. The first real number we would have.
5. An evidence-based answer on flight altitude, currently unsettled.
6. A possible publication: tiling, aerial person detection and low-power onboard inference have not been combined for flood conditions.
7. Code and results published either way. A negative result settles the same question and costs the same.

5. Benefits

Operational. Three aircraft sweep ten hectares in under eight minutes and hand a coordinator a list of coordinates rather than a video feed somebody must watch — with no network, which is the condition that actually obtains.

Economic. The whole system costs less than one hour of helicopter search time, travels by road, and is run by two people.

Departmental. What outlives the project: a calibrated thrust stand, a certified scale, a ground station, a tested autonomy codebase, and students who have taken a system from arithmetic to flight.

Strategic. Indian suppliers wherever they meet the specification — for sovereignty, and because domestic lead times are half those of imports, which buys flight-test days.

Transferable. Nothing is flood-specific except the training data. The same system suits earthquake rubble survey, wildfire perimeter mapping and avalanche search.

6. Resources needed

The order is engineering, not accounting: prove the motor before buying twelve, and fly one aircraft before building three. **No step exceeds INR 30,000**, and each is released only when the previous one has produced a result — so the department is never committed to more than one small step.

#	Step	INR	Gate
1	Thrust stand + calibrated scale	26,196	On approval
2	**One motor**, to measure	5,175	Stand built
Aircraft one — released only on measured thrust ≥ 3.18 kgf per motor			
3	Flight controller	25,990	Thrust proven
4	Onboard computer + AI accelerator	21,850	—
5	GNSS receiver (centimetre-accurate)	27,892	—
6	Camera, radio, mesh	29,440	—
7	Airframe (carbon tube, clamps)	20,144	Avionics talking
8	Battery cells + pack assembly	29,299	—
9	Speed controllers, release, propellers, wiring	24,223	—
10	Remaining three motors	15,525	—
11	Charger + safety-pilot radio	25,300	Airframe weighed
12	Consumables + sun hood	17,274	—
Ground station — released on first flight			
13	Base station receiver	27,892	First flight
14	Survey tripod	4,600	—
15	Field cables, mounts, spares	28,750	—
16	Registration and remaining statutory	5,750	—
Aircraft two — released only after aircraft one flies a full mission			
17–24	Same eight steps as 3–10	1,99,538	Mission flown
Aircraft three — released only after two aircraft fly with verified separation			
25–32	Same eight steps as 3–10	1,99,538	Separation verified
	Total	**7,34,379**	

Steps 17–32 repeat steps 3–10 exactly, once per aircraft, none above INR 29,440. By the time any is requested a complete aircraft has already flown — so what is funded is **replication of a working design**. (Individual steps are rounded to the rupee, so the column sums to within a few rupees of the total.)

Named parts. Holybro Pixhawk 6C Mini; Raspberry Pi 5 8 GB with the Pi AI HAT+ (Hailo-8); Arducam IMX477 with a 6 mm *fixed-focus* lens (hyperfocal distance is 4.15 m, so focus set once is correct at every altitude we fly); Molicel P45B cells, 4500 mAh at 45 A because peak draw is 38.3 A per cell; Tarot TL96020 motors; ToolkitRC M6D chargers; RadioMaster Boxer.

Two notes. The obvious charger, the HOTA D6 Pro at INR 10,499–INR 13,500, is 325 W per channel and can barely charge our packs; the ToolkitRC M6D is 500 W at INR 5,836 and needs only a power supply the department already owns. And one line is *under*-budgeted: propellers are carried at INR 1,000 each against a verified INR 3,250 per pair — about INR 7,500 short across three aircraft, on the item most often broken. We will seek a cheaper supplier and absorb the rest from contingency.

What this costs. Thirty-two approvals is real administrative work, and buying one aircraft at a time means placing the same imported order three times — roughly eight to twelve weeks of schedule. We have chosen it anyway, because we would rather move slowly with your confidence than quickly without it. If either becomes binding, merging the per-aircraft steps is the lever, and that is your call.

For the detection test we need nothing new: about a week on a departmental GPU and 30 GB of storage. The dataset is public, the software open source.

7. Future work

Near term. Replace every modelled number with a measured one — detection accuracy, motor thrust, setup time. Test near-infrared sensing: water absorbs infrared and people do not, so a person appears bright against dark water. A camera without its infrared filter costs about INR 3,000 and a public dataset exists to test it before buying. Beyond floods, the same system suits earthquake, wildfire and avalanche search.

Longer term: aircraft that must think for themselves. Our central constraint — a capable aircraft deciding for itself inside a very small power budget — is a hard requirement in planetary aviation. NASA's *Ingenuity* flew 72 times on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth's density. *Dragonfly* launches in 2028 for Titan, where the air is four times denser than Earth's at one seventh the gravity.

What makes those hard is not the flying but that **a radio command takes minutes to arrive**: nobody is piloting, so every choice is made on board. That is our problem with the difficulty raised, and what transfers is what is not about floods — planning a survey on board, running perception to an *energy* budget, and failing safely with no link.

We should not overstate it. Our own analysis found that microcontroller-class processors — the hardware usually proposed for such missions — cannot see our targets at all, because shrinking the image to fit them destroys what we are looking for. A planetary version needs a different operating point. That is a research question, not a slide.

8. Conclusion

The design is further along than it looks, and explicit about which numbers are measured and which are not. We are asking for the means to finish it in 32 small steps, none above INR 30,000, so nothing is committed before the previous step has produced evidence.

The first step is INR 26,196 and answers the question everything depends on: can these motors lift this aircraft. The immediate technical need is smaller still — a week of GPU time to test the detection idea on free public data. Having already found one sizing mistake by checking, testing first seems clearly the better choice.

We would value your guidance on four decisions that are ours but that we would rather not make alone: the body size we design against, the operating altitude, whether to adopt the two-stage detector, and which of two processing modules to buy.